

Education

University of Science and Technology of China

Sep 2024 - Jun 2027 (expected)

Master's study, Intelligent Science and Technology; School of Artificial Intelligence and Data Science

Hefei University of Technology

Sep 2020 - Jun 2024

Undergraduate study, Information Security; School of Computer Science and Information Engineering

Research interests

Agentic engineering; 3D generation; autoregressive modeling; computer graphics; AI-generated content (AIGC)

Research and projects

Towards Valid B-Rep Generation: Training-Free Wireframe Anomaly Detection and Repair

First author; submitted to CVM, decision pending

Dec 2025 - May 2026

Preprint: arXiv:2608.04955

- Proposed training-free WDR, combining GTAD anomaly detection and EGGTR energy-guided repair at the wireframe stage. Detects and routes self-intersections, collapsed edges and disconnected vertices for local regeneration; supports autoregressive and diffusion B-Rep generators without retraining the base model.
- Integrated WDR into DTGBrepGen, Stitch-A-Shape and BrepForge. On 3,000 paired samples, improved DTGBrepGen validity from 79.5% to 95.6% on DeepCAD and from 68.1% to 83.9% on ABC, largely preserving generation diversity and distribution quality.

Stack: Python, PyTorch, Diffuser, PythonOCCT

TrussOrbit: Edge-Orbit Sequence Modeling for Truss Metamaterial Inverse Design

First author; submitted to ICLR 2027, decision pending

Jul 2026 - Sep 2026

- Proposed EOTR to jointly encode node geometry and connectivity through edge-orbit sequences. Compact sequences and symmetry expansion reconstruct full trusses for target-stiffness-driven autoregressive inverse design.
- Proposed set-level reinforcement learning with SMCO: finite-element feedback and empty-slot counterfactual evaluation assign sequence-level marginal credit, combined with GSPO to jointly optimize stiffness matching and physical validity.
- On a unified benchmark of 1.49 million structures built by our team, outperformed all compared baselines in speed and reconstruction error for single-shot inverse generation from stiffness matrices, while supporting a larger topology design space.

Stack: Python, PyTorch, VeRL, SGLang

Automated research system based on coding agents

Project lead and core contributor

Sep 2026 - Present

- Built the framework and led Idea Proposal and automated experiments. Implemented customizable research ideation with the Codex SDK, integrating within- and cross-domain literature, prior experiments and researcher input for iterative human-agent collaboration.
- Produced an Idea Package containing research gaps, method concepts, introduction figures, baselines, datasets and metrics. Used it to select experiment types and templates, plan execution, and schedule and monitor experiments through Codex.
- Consolidated experimental processes and results into an Experimental Package as evidence for downstream paper-generation modules, connecting ideation, execution and results delivery.

Stack: Codex SDK, Python

Research and projects (continued)

Autoregressive wireframe generation

Project lead

- Led the team in the CSIAM GDC 2026 CAD Wireframe Neural Compression Challenge: EdgeGPT ranked 1st in the preliminary round; CurveGPT ranked 2nd in the final. Competition website
- Evolved EdgeGPT into CurveGPT with GPT-style autoregressive decoding and a constrained, typed-edge grammar for LINE, ARC and COMPLEX curves. Designed HQVAE to compress complex curves into discrete RQ tokens and constrained generation with an inference state machine to reduce format failures in long sequences.
- Cleaned about 450,000 records from the official and ABC datasets; built three-stage training and a complete evaluation pipeline.

Stack: Python, PyTorch, Transformers

Design and implementation of a tiled software rasterization pipeline

Undergraduate thesis

Dec 2023 - May 2024

- Independently implemented a programmable CUDA software rasterizer with Triangle Setup, Bin Raster, Coarse Raster and Fine Raster stages, primitive-order preservation and an interactive rendering system.
- Refactored warp-level synchronization and thread-safe interfaces for correctness and scalability on modern GPUs. Proposed CPU-scheduled multi-pass 4K rendering to overcome previous resolution and concurrency limits.
- Used dynamic shared-memory allocation, reduction optimization, timer instrumentation and Nsight Compute to identify binning/coarse-raster bottlenecks and derive near-optimal block configurations for the architecture.

Stack: C++, OpenGL, CUDA

Privacy-preserving navigation with unordered waypoints on encrypted urban maps

Fourth author of the resulting paper; competition project lead

Dec 2022 - Jun 2023

- Contributed to Semi-Constrained Navigation, with results published in IEEE TDSC (CCF-A): navigation through k unordered waypoints on encrypted urban maps, balancing accuracy, privacy and query efficiency.
- Proposed precomputation-based acceleration, improving the execution efficiency of key homomorphic-encryption operators by 3–6 $\times$; contributed to the path-distance oracle, encrypted queries and cloud-side routing logic.
- Built the backend interaction pipeline for key management, ciphertext communication and query services using Java, JPBC and Spring Boot; validated real-time performance and usability at urban-map scale.

Stack: Java, JPBC, Spring Boot

Awards and honors

- Jul 2026: 2nd place, CSIAM GDC 2026 CAD Wireframe Neural Compression Challenge.
- Nov 2024: Outstanding Undergraduate Thesis, Anhui Computer Federation (provincial).
- Jun 2024: Outstanding Graduate, Hefei University of Technology (university).
- Dec 2022: National First Prize, C++ division, National College Computer Ability Challenge.
- Dec 2022: National Scholarship (national).
- Jul 2022: Third Prize, undergraduate programming track, Anhui Robot Competition (provincial).

Skills

Foundations: Data structures and algorithms, computer networks, operating systems and GPU architecture for model training, systems development and performance optimization.

Programming: Python, C/C++, CUDA and Java; independent algorithm development, implementation and debugging.

Training tools: PyTorch, Transformers and DeepSpeed; training pipelines and distributed-training acceleration.

AI-assisted workflows: LLM-assisted coding, literature research and technical understanding to support development and research iteration.